MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746256
CONTROL PANEL-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of Specification MIL-C-48411 comprise the complete requirements for Control Panel-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assembly, General Specification for

DRAWINGS

11746256	Circuit Card Assembly, Control Panel-2

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Identification.- This assembly is the second of two (2) single ended printed circuit boards consisting of gates, flip-flops, and inverters necessary to provide the logic circuitry to light the various lamps located on the front panel of the Simulated Target Generator.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746256.

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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +1 VDC
b. +5 VDC
c. +10 VDC
d. +15 VDC
e. +28 VDC
f. -15 VDC
g. 0 to +1 VDC Variable
h. 0 to +10 VDC Variable

3.2.1.2 Voltage Divider.- The output at pin 33 shall be 7.5 VDC ± .5 VDC.

3.2.1.3 Logic light switches.- The outputs at Pins 12, 16, 18, 28, 35, and 37 shall be capable of energizing and deenergizing their respective lamps.

3.2.1.4 Test Voltages.- The test voltages for pins 9, 13, 15, and 29 shall be as shown in Tables 4, 5, 6, and 10 respectively.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746256 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746256 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

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4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment.- (see Figure 1)

4.2.1.1 Power supply (PS1).- The power supply shall be 5 VDC ± 1 mVDC, 100 A or greater and a ripple of less than 10 mVRMS.

4.2.1.2 Power supply (PS2).- The power supply shall be 10 VDC ± 1 mVDC, 100 mA or greater and a ripple of less than 10 mVRMS.

4.2.1.3 Power supply (PS3).- The power supply shall be 15 VDC ± 1 mVDC, 100 mA or greater and a ripple of less than 10 mVRMS.

4.2.1.4 Power supply (PS4 & PS5).- The power supplies shall be 20 VDC ± 1 mVDC, 100 mA or greater and a ripple of less than 10 mVRMS.

4.2.1.5 Power supply (PS6).- The power supply shall be 40 VDC ± 1 mVDC, 100 mA or greater and a ripple of less than 10 mVRMS.

4.2.1.6 Power supply (Variable) (PS4).- The power supply shall be adjustable with an output of ± 1 mV for any setting of from 0 to 15 VDC, 100 mA or greater, a ripple of less than 500 VDC and a load effect of 0.01% + 1 mV.

4.2.1.7 Oscilloscope.- Rise time of scope and associated measuring probe of 50 nanoseconds of less, sweep of 50 nanoseconds per centimeter or greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.8 Signal generator.- The generator shall be capable of a 7.5 Hz ± .5 Hz, 0.00 to 5.00 V, and 10 nanaseconds rise and fall time square wave output.

4.2.1.9 Dummy Loads.- The dummy loads shall be MS 25237-327 miniture lamps or equivalent, and a 3-input positive 'hand' gate type SN 5410 or its equivalent.

4.2.1.10 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

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4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

a. +1.000 VDC ± 1 mVDC
b. +5.000 VDC ± 1 mVDC
c. +10.000 VDC ± 1 mVDC
d. +15.000 VDC ± 1 mVDC
e. +28.000 VDC ± 1 mVDC
f. -15.000 VDC ± 1 mVDC
g. 0.000 VDC to + 1.000 VDC ± 1 mVDC Variable
h. 0.000 VDC to +10.000 VDC ± 1 mVDC Variable

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC, the "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.4 Voltage Divider.- When +15.00 VDC is applied to the circuit board the output at pin 33 shall be 7.5 VDC ± .5 VDC.

4.2.2.5 Logic Light Switches.- With the circuit board connected as shown in Figure 1, the inputs listed in Tables below shall yield the corressponding outputs.

4.2.2.6 Test Voltages.- With the circuit board connected as shown in Figure 1, and with the inputs listed in Tables 4,5,6, and 10 shall yield the corressponding outputs.

TABLE 1

	Input (VDC/Logic)			Output
Pins	27	32	34	37(L1)
	Logic "1"	+28VDC	Logic "0"	ON
	Logic "1"	+28VDC	+28VDC	OFF

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TABLE 2

Input (VDC/Logic)	Output
Pin 32	Pin 35(L2)
+28VDC	ON
Logic "0"	OFF

TABLE 3

Input (VDC/Logic)		Output
Pin 32	Pin 27	Pin 12(L3)
+28VDC	Logic "0"	OFF
+28VDC	Logic "1"	ON

TABLE 4

Input (VDC/Logic)		Output
Pin 27	Pin 32	Pin 13
Logic "0"	+28VDC	+15.0VDC ± .1VDC
Logic "1"	+28VDC	gnd.

TABLE 5

Input (VDC/Logic)		Output
Pin 27	Pin 32	Pin 15
Logic "0"	+28VDC	-6.43VDC ± 30MVDC
Logic "1"	+28VDC	+10VDC ± .1VDC

TABLE 6

Input (VDC/Logic)		Output
Pin 27	Pin 32	Pin 9
Logic "0"	+28VDC	+6.43VDC ± 30MVDC
Logic "1"	+28VDC	+1.6VDC ± 60MVDC

TABLE 7

a. Apply the variable voltage capable of going from -1 VDC to +1 VDC to Pin 40.
b. Apply Logic "0" to Pin 22.

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TABLE 7 CON'T

c. Apply Logic "0" to Pin 23.
d. Reset "0" on Pin P14.
e. Vary the voltage incrementally on Pin 40 from -1 VDC to +1 VDC.
f. When the voltage on Pin 40 has exceeded 0 VDC by .1 VDC or less lamp L4 (Pin 16) shall light and remain on.
g. Apply 1 VDC $\pm$ .1 VDC to Pin 40.
h. Apply Logic "1" to Pin 23.
i. Apply Logic "1" to Pin 22.
j. Lamp L4 (Pin 16) shall blink.

TABLE 8

a. Apply the variable voltage capable of going from -1 VDC to +1 VDC to Pin 40.
b. Apply Logic "0" to Pin 23.
c. Reset "0" on Pin 14.
d. Vary the voltage incrementally on Pin 40 from -1 VDC to +1 VDC.
e. When the voltage on Pin 40 has exceeded 0 VDC by .1 VDC or less lamp L5 (Pin 18) shall light and remain on.
f. Apply 1 VDC $\pm$ .1 VDC to Pin 40.
g. Apply Logic "1" to Pin 25.
h. Apply Logic "1" to Pin 30.
i. Apply Logic "1" to Pin 23.
j. Lamp L4 (?) shall blink.

TABLE 9

a. Apply +1 VDC $\pm$ .1 VDC to Pin 40.
b. Apply Logic "0" to Pin 10.
c. Apply Logic "0" to Pin 31.
d. Lamp L6 (Pin 28) shall light.
e. Apply Logic "1" to Pin 10.
f. Lamp L6 (Pin 28) shall go off.
g. Vary the voltage incrementally on Pin 31 from -10 VDC to +10 VDC.
h. When the voltage on Pin 31 has reached +6.25 VDC $\pm$ .1 VDC Lamp L6 (Pin 28) shall light and remain on.

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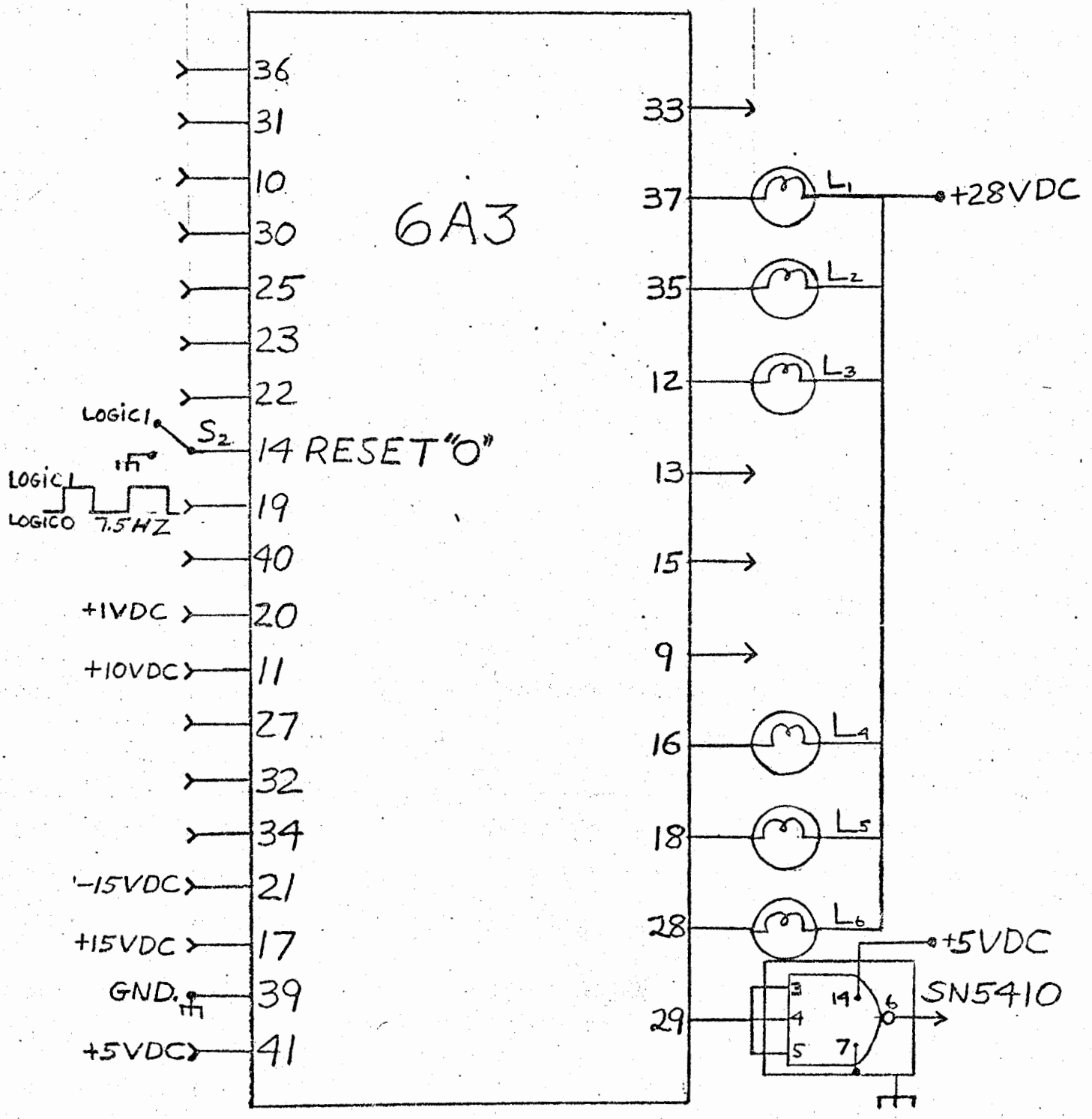


FIGURE 1

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TABLE 10

Input				Output
Pin 10	Pin 36	Pin 40	Pin 31	Pin 29
Logic "1"	Logic "1"	+1VDC ± .1VDC	Logic "0"	Logic "1"
Logic "1"	Logic "0"	+1VDC ± .1VDC	Logic "0"	Logic "0"
Logic "0"	Logic "1"	+1VDC ± .1VDC	Logic "0"	Logic "0"
Logic "0"	Logic "1"	+1VDC ± .1VDC	+10VDC ± .1V	Logic "0"

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746256.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applied to this specification.

Custodian:

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Preparing Activity:

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Project No. 1240-A598

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